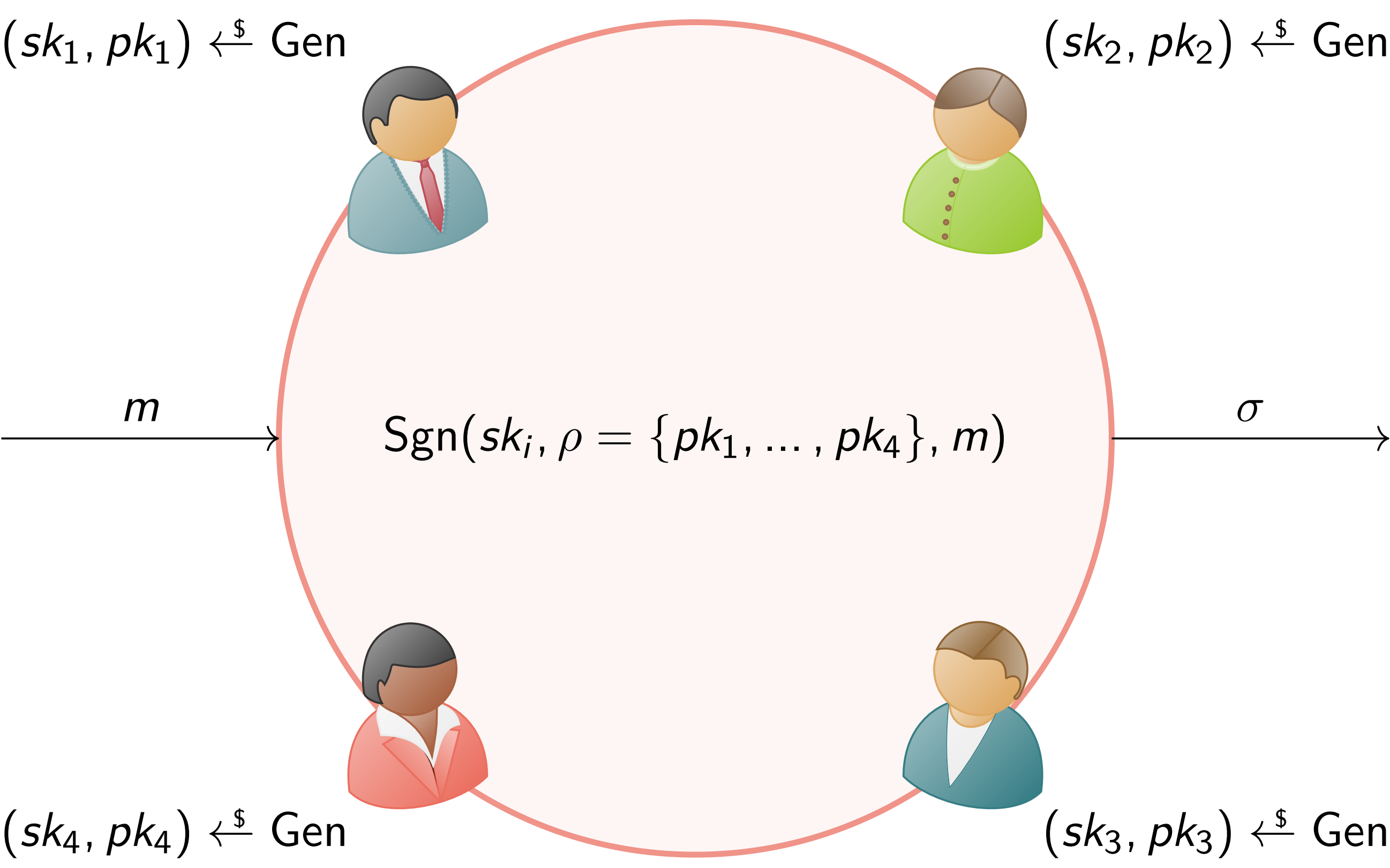
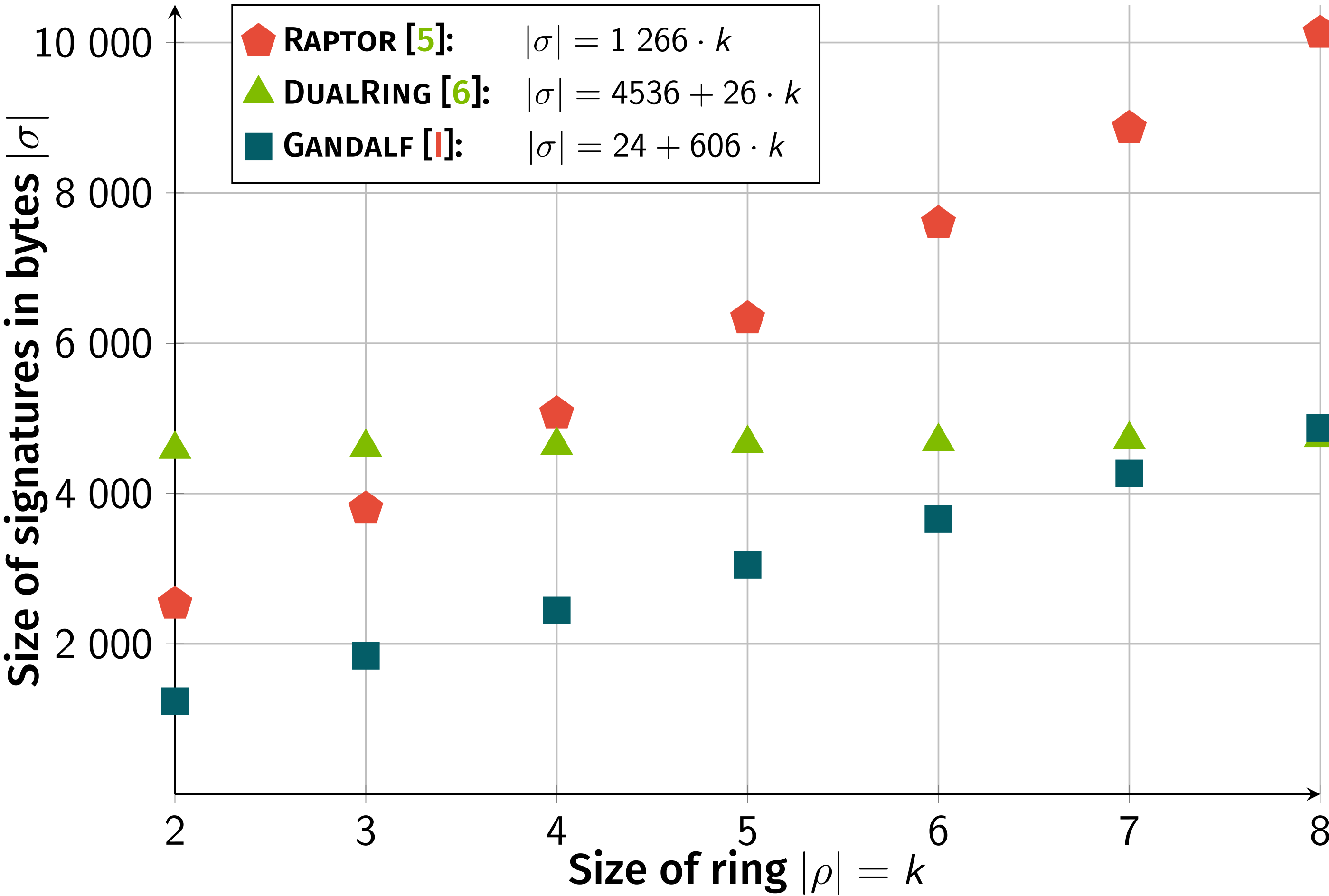




Ring Signature Scheme — RSig = (Gen, Sgn, Ver)



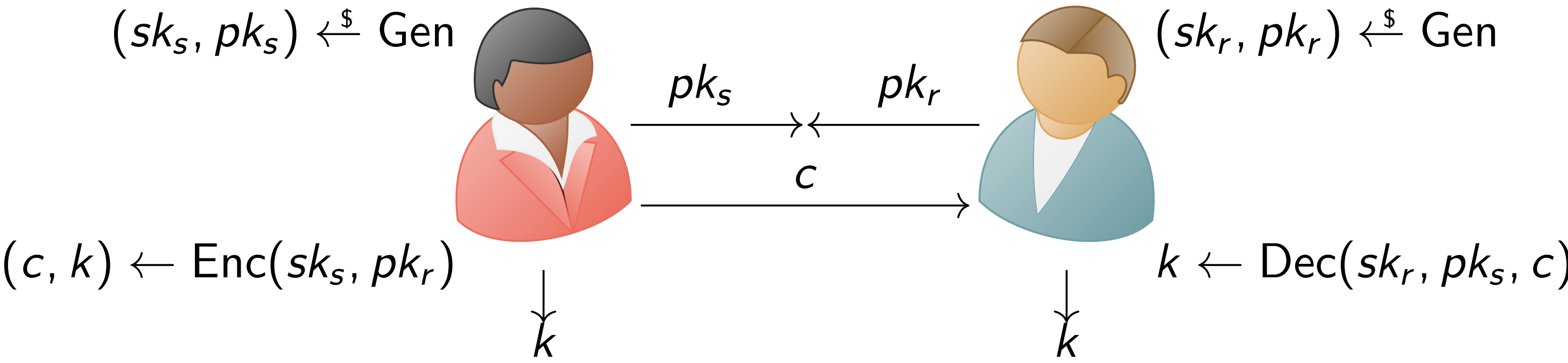
- Ring signatures allow users to sign messages on behalf of dynamically formed groups, and are publicly verifiable while preserving the signer's anonymity within the group (signing ring ρ).
- Ring signatures are widely deployed in blockchains and voting systems.
- We construct a ring signature scheme, GANDALF [I], specifically for small rings, providing 50% reduction in signature sizes over the state of the art.



- GANDALF, is based on the NTRU pre-image sampleable trapdoor function f_h over the NTRU ring.
- Concretely, f_h inputs two ring elements of small norm and is defined as $f_h(u, v) = h * u + v$. A valid ring signature on message m for the ring ρ consists of a vector $(u_1, \dots, u_k) \in \mathcal{R}^k$ such that

$$\left\| \left(u_1, \dots, u_k, v = H(m, \rho = \{h_1, \dots, h_k\}) - \sum_{i=1}^k h_i * u_i \right) \right\|_2 \leq \beta.$$

Deniable Authenticated Key Exchange Mechanisms — AKEM = (Gen, Enc, Dec)



- An AKEM can be thought of the KEM analogue of signcryption and was first formalised in [7]. It is the primitive behind the HPKE [8] standard used in MLS and TLS.
- Our work [I] introduces and formalises deniability for AKEMs.
- Furthermore, we show a black box construction of a deniable AKEM using our ring signature scheme GANDALF.

- We present the first black-box construction of a deniable AKEM combiner [III] that retains statistical rather than computational deniability.
- Using GANDALF, we instantiate a concrete combiner SHADOWFAX, with small ciphertexts and public keys, and competitive performance.

Scheme (variant)	Confidentiality	Authenticity	Deniability	Assumption		Size (in bytes)	
				pre-Q	post-Q	c	pk
DH-AKEM (Curve25519) [7]	Ins-CCA	Out-Aut	DR*	✓	✗	32	32
EtStH-AKEM (BAT + ANTRAG) [9]			X	✗	✓	1 119	1 417
NIKE-AKEM (SWOOSH) [9]			DR*	✗	✓	> 221 184	> 221 184
FrodoKEX+ [10]	IND-1BatchCCA	UNF-1KCA	DR	✗	✓	72	21 300
PQ-AKEM (NTRU-A + GANDALF) [I]	Ins-CCA	Out-Aut	HR & DR	✗	✓	2 044	1 664
PQ-AKEM (BAT + GANDALF) [I]				✗	✓	1 749	1 417
SHADOWFAX (Curve25519 + BAT + GANDALF) [III]				✓	✓	1 781	1 449

Performance

Cycle counts (in kilocycles) and time (in milliseconds) for various AKEMs and ring signature schemes on a 3GHz Firestorm core of an Apple M1 Pro. For the post-quantum AKEM (PQ-AKEM) from [I], we instantiate the underlying KEM with bat_257_512.

AKEM	Gen		Enc		Dec	
	kcc	ms	kcc	ms	kcc	ms
DH-AKEM [7]	208	0.07	628	0.21	419	0.14
PQ-AKEM [I]	24 748	8.25	1215	0.41	332	0.11
SHADOWFAX [III]	24 770	8.27	1846	0.62	746	0.25
RSig	Gen		Sgn		Ver	
	kcc	ms	kcc	ms	kcc	ms
Raptor [5]	71 420	23.81	7 980	2.66	505	0.17
Gandalf [I]	13 145	4.38	1 110	0.37	95	0.03

Select References

[I] P. Gajland, J. Janneck, E. Kiltz. *Ring Signatures for Deniable AKEM: Gandalf's Fellowship*. CRYPTO 2024.

[II] P. Gajland, V. Hwang, J. Janneck. *Shadowfax: A Practical Deniable AKEM Combiner*. Submitted to USENIX 2025.

[4] T. Prest, P.-A. Fouque, J. Hoffstein, P. Kirchner, V. Lyubashevsky, T. Pornin, T. Ricosset, G. Seiler, W. Whyte, Z. Zhang. *FALCON*. Technical report, National Institute of Standards and Technology, 2022.

[5] X. Lu, M. H. Au, Z. Zhang. *Raptor: A Practical Lattice-Based (Linkable) Ring Signature*. ACNS, 2019.

[6] T. H. Yuen, M. F. Esgin, J. K. Liu, M. H. Au, Z. Ding. *DualRing: Generic Construction of Ring Signatures with Efficient Instantiations*. CRYPTO, 2021.

[7] J. Alwen, B. Blanchet, E. Hauck, E. Kiltz, B. Lipp, D. Riepel. *Analysing the HPKE Standard* EUROCRYPT, 2021.

[8] R. Barnes, K. Bhargavan, B. Lipp, C. A. Wood. *Hybrid Public Key Encryption*. RFC 9180, 2022.

[9] J. Alwen, J. Janneck, E. Kiltz, B. Lipp. *The Pre-Shared Key Modes of HPKE* ASIACRYPT, 2023.

[10] D. Collins, L. Huguenin-Dumittan, N. K. Nguyen, N. Rolin, S. Vaudenay. *K-Waay: Fast and Deniable Post-Quantum X3DH without Ring Signatures*. USENIX, 2024.